

Multiple Choice (10 marks)

1. If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area does which of the following?
 - (a) Decreases by a factor of 81.
 - (b) Decreases by a factor of 9.
 - (c) Increases by a factor of 9.
 - (d) Increases by a factor of 81.
2. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.
3. An atom has filled $n = 1$ and $n = 2$ levels. How many electrons does the atom have?
 - (a) 2
 - (b) 4
 - (c) 6
 - (d) 10
4. If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is
 - (a) $mc/2$
 - (b) $mc/(2^{1/2})$
 - (c) mc
 - (d) $(3^{1/2})mc$
5. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
 - (a) $0.1 \text{ GeV}/c^2$
 - (b) $0.2 \text{ GeV}/c^2$
 - (c) $0.5 \text{ GeV}/c^2$
 - (d) $1.0 \text{ GeV}/c^2$

True / False (10 marks)

Answer True or False

6. True or False: The attraction between Cooper pairs in superconductors is primarily due to the strong nuclear force.
7. True or False: Electromagnetic radiation emitted from a nucleus is most commonly in the form of gamma rays.
8. True or False: In a single-electron atom with $l = 2$, there are only 3 allowed values for the magnetic quantum number m_l .
9. True or False: The negative muon, μ^- , *has properties most similar to mesons, which are composite particles.*
9. True or False: Gas lasers utilize transitions that involve the energy levels of free atoms to produce laser light.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the internal energy of an ideal gas changes during a constant volume process, specifically analyzing the relationship between heat added and internal energy increase.
12. Discuss the allowed transitions for an electron in the $n = 4, l = 1$ state of hydrogen, and analyze which final states are not achievable, providing reasoning for your conclusions.

End of Paper